

Entitlement Pipeline: Data Cleaning

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1 Goal and Plan

The goal of this project at this point is to map the full housing construction pipeline. How many units are proposed under the entitlements system, how many get approved and how many get built. The crucial elements of this are:

1. Having a project level file. There are multiple entitlements for the same project. We need to be confident that we have a single row for each project. How should we handle an entitlement that was proposed, not built and then they try again. If it was at least five years later or is a different kind of project.

Key Variables We Need:

- Date of proposal, completion
- Location
- What they were asking for: units, free text field, type of entitlement
- Level of opposition: appeals, environmental review

2 A Project Level Entitlements File

We want there to be a single row for every proposed entitlement project in the data. Some questions:

1. How can we link across cases? There are APNs and addresses on each project that allow us to link across cases. If we make some mistakes, it seems okay as long as we don't lead to missing merges because of the different APNs.
2. What if there are multiple APNs for the same project? This is the same issue as above.

3. What if there are multiple distinct projects per APN? Like someone tries something now and they also had tried five years ago. I would call that a failure and then a success, so I would want to have like five year buckets.
4. How can we validate this data?

2.1 Tasks

1. Clean Post 2015 File
2. Clean Pre 2015 File and assess how data quality is different across the two
3. Handle duplicates
4. Classify type of entitlements
5. Summary statistics

2.2 Data Sources

LA entitlements provides a handful of files. In the pre and post 2015 data, each row is a case. This means that there are environmental, and appeals cases that are tied to a main entitlement case all combined. The first step in cleaning the data is to separate out appeals and environmental cases to be wide on the main case. This lets us observe the full length of time for this cases but

2.3 Going from Cases to Applications

Okay so at this point our data is wide where each row is a case and we observe the environmental review case and the appeals case wide. We know that there are many such cases within a single application. How should we aggregate information?

- We would want to know if any of the cases are denied
- We would want to know the full length associated with the application
- We would want the total hold days across the different cases
- We would want to be sure there was no geographic differences across the cases. We would want to know what all the entitlements asked for are.

2.4 Classifying the Type of Entitlement

Different entitlements have different levels of heavy lift.

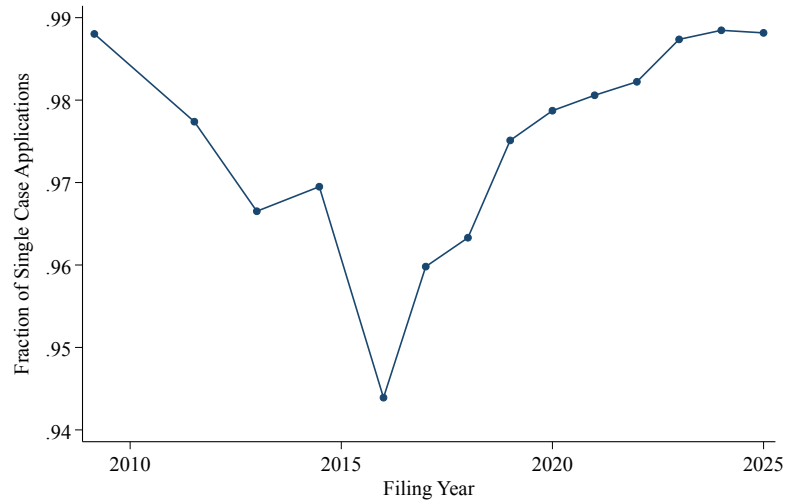


Figure 1: Fraction of Applications with Single Case

3 A Clean Building Permits Database

4 Linking Entitlements to Building Permits